



Lab 02: Cloud Storage Qwik Start - CLI/SDK

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Overview

Same Cloud Storage concepts as Lab 01 but done entirely through the command line using gcloud storage commands in Cloud Shell instead of the GCP Console. This lab reinforces that most GCP tasks can be done both via UI and CLI.

Lab 01 vs Lab 02 - Console vs CLI

Lab 01 - Console	Lab 02 - CLI
Point and click in browser	gcloud storage commands
IAM roles via Permissions tab	ACL grants via --add-acl-grant
Upload via file dialog	Upload via gcloud storage cp
Cannot automate	Can be scripted and automated

Objectives

- Create a Cloud Storage bucket via CLI
- Upload an object to the bucket via CLI
- Download an object from the bucket
- Copy an object to a folder in the bucket
- List contents of a bucket or folder
- List details for an object
- Make an object publicly accessible via CLI
- Remove public access from an object

Setup - Set Region

```
gcloud config set compute/region [REGION]
```

```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try p
Your Cloud Platform project in this session is set to qwiklabs-gcp-00-19d4a49a
Use `gcloud config set project [PROJECT ID]` to change to a different project
student_01_c608d40bd029@cloudshell:~ (qwiklabs-gcp-00-19d4a49a98be)$ gcloud co
Updated property [compute/region].
```

Task 1 - Create a Bucket

```
gcloud storage buckets create gs://[YOUR-BUCKET-NAME]
```

☐	Name ↑	Created	Location type	Location	Default storage class ⓘ	Last modified	Public access ⓘ	Access contro
☐	niyah_bucket	May 20, 2026, 11:11:02 PM	Multi-region	us	Standard	May 20, 2026, 11:11:02 PM	Subject to object ACLs	Fine-grained ⋮

Task 2 - Upload an Object

Downloaded the Ada Lovelace portrait image from Wikipedia then uploaded it to the bucket.

```
curl
https://upload.wikimedia.org/wikipedia/commons/thumb/a/a4/Ada_Lovelace_portrait.jpg
/800px-Ada_Lovelace_portrait.jpg --output ada.jpg
gcloud storage cp ada.jpg gs://[YOUR-BUCKET-NAME]
rm ada.jpg
```

Create folder

Upload ▾

Transfer data ▾

Create batch operations

New

Other services ▾

Learn

Filter by name prefix only ▾

Filter

Filter objects and folders

Show

Live objects only ▾

☐	Name	Size	Type	Created ?	Storage class	Last modified	Object contexts ?	Publ	
☐	 ada.jpg	2 KB	image/jpeg	May 20, 2026, 11:17:14 PM	Standard	May 20, 2026, 11:17:14 PM	—	Not	 

Task 3 - Download an Object

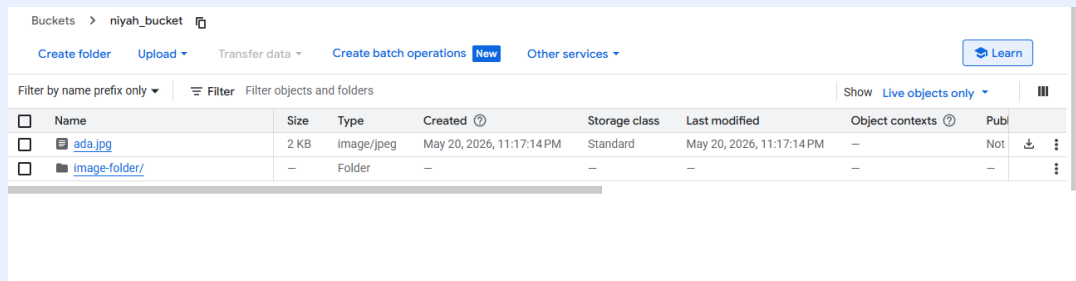
```
gcloud storage cp -r gs://[YOUR-BUCKET-NAME]/ada.jpg .
```

```
student_01_c608d40bd029@cloudshell:~ (qwiklabs-gcp-00-19d4a49a98be)$ rm ada.jpg
student_01_c608d40bd029@cloudshell:~ (qwiklabs-gcp-00-19d4a49a98be)$ gcloud storage cp -r gs://niyah_bucket/ada.jpg .
Copying gs://niyah_bucket/ada.jpg to file:///ada.jpg
Completed files 1/1 | 2.0kiB/2.0kiB
student_01_c608d40bd029@cloudshell:~ (qwiklabs-gcp-00-19d4a49a98be)$
```

Task 4 - Copy Object to a Folder

Created an image-folder inside the bucket and copied ada.jpg into it.

```
gcloud storage cp gs://[YOUR-BUCKET-NAME]/ada.jpg gs://[YOUR-BUCKET-NAME]/image-
folder/
```



☐	Name	Size	Type	Created	Storage class	Last modified	Object contexts	Public	
☐	ada.jpg	2 KB	image/jpeg	May 20, 2026, 11:17:14 PM	Standard	May 20, 2026, 11:17:14 PM	—	Not	
☐	image-folder/	—	Folder	—	—	—	—	—	

Task 5 - List Contents of a Bucket

```
gcloud storage ls gs://[YOUR-BUCKET-NAME]
```

```
Completed files 1/1 | 2.0kiB/2.0kiB
student_01_c608d40bd029@cloudshell:~ (qwiklabs-gcp-00-19d4a49a98be)$ gcloud storage ls gs://niyah_bucket
gs://niyah_bucket/ada.jpg
gs://niyah_bucket/image-folder/
student_01_c608d40bd029@cloudshell:~ (qwiklabs-gcp-00-19d4a49a98be)$
```

Task 6 - List Details for an Object

```
gcloud storage ls -l gs://[YOUR-BUCKET-NAME]/ada.jpg
```

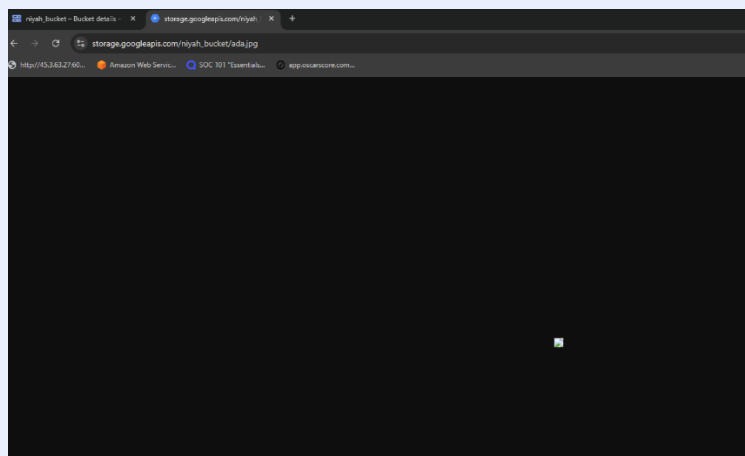
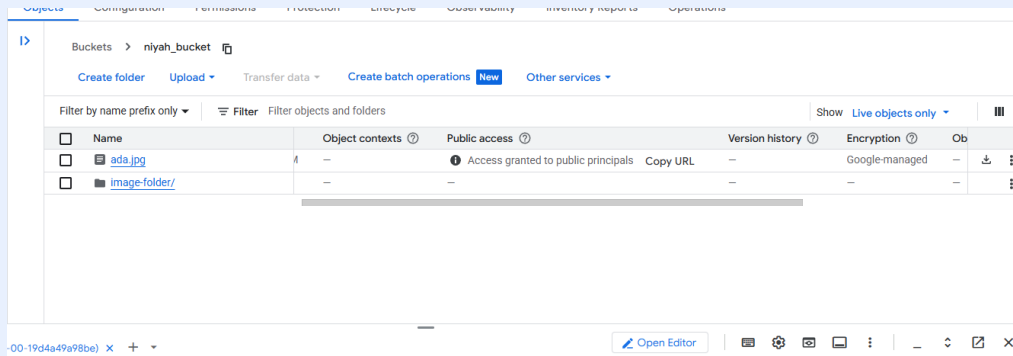
```

Completed files 1/1 | 2.0kiB/2.0kiB
student_01_c608d40bd029@cloudshell:~ (qwiklabs-gcp-00-19d4a49a98be)$ gcloud storage ls gs://niyah_bucket
gs://niyah_bucket/ada.jpg
gs://niyah_bucket/image-folder/
student_01_c608d40bd029@cloudshell:~ (qwiklabs-gcp-00-19d4a49a98be)$ gcloud storage ls -l gs://niyah_bucket/ada.jpg
2009 2026-05-21T03:17:14Z gs://niyah_bucket/ada.jpg
TOTAL: 1 objects, 2009 bytes (1.96kiB)
student_01_c608d40bd029@cloudshell:~ (qwiklabs-gcp-00-19d4a49a98be)$

```

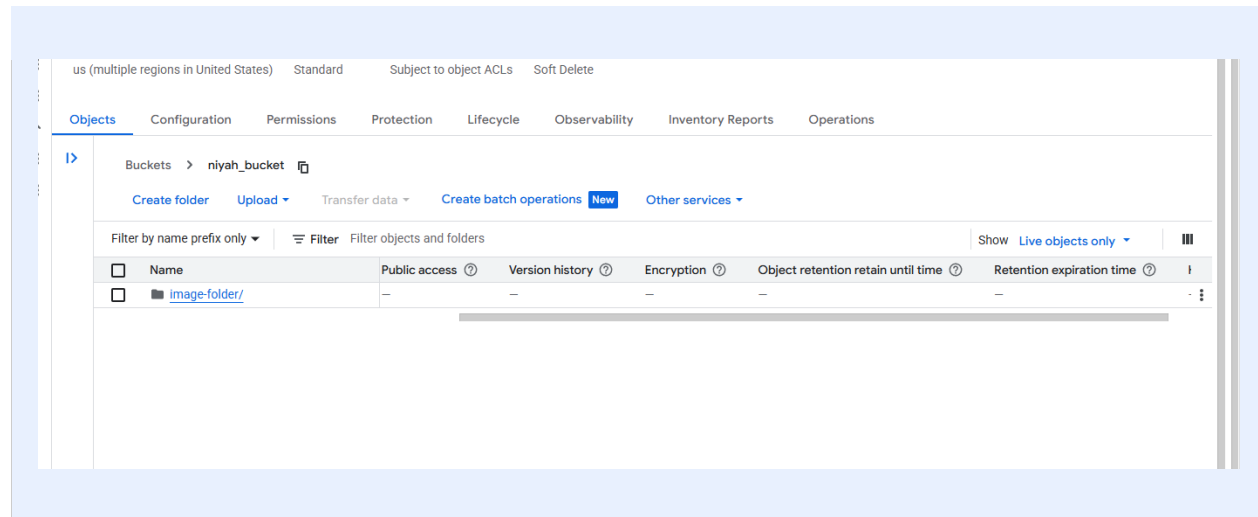
Task 7 - Make Object Publicly Accessible

```
gcloud storage objects update gs://[YOUR-BUCKET-NAME]/ada.jpg --add-acl-grant=entity=allUsers,role=READER
```



Task 8 - Remove Public Access and Delete

```
gcloud storage objects update gs://[YOUR-BUCKET-NAME]/ada.jpg --remove-acl-grant=allUsers
gcloud storage rm gs://[YOUR-BUCKET-NAME]/ada.jpg
```



Real World Application

Using CLI for Cloud Storage is preferred in real jobs because:

- Scripts can automate uploading thousands of files at once
- CI/CD pipelines use gcloud commands to deploy assets automatically
- Faster than clicking through the console for repetitive tasks
- Can be combined with other shell commands for powerful workflows

What I Learned

- gcloud storage cp uploads and downloads objects from buckets
- gcloud storage ls lists bucket contents and object details
- Public access is granted via --add-acl-grant=entity=allUsers,role=READER
- Public access is removed via --remove-acl-grant=allUsers
- CLI can do everything the Console can do and more
- Folders in Cloud Storage are virtual paths in object names

Challenges & How I Solved Them

Named the bucket with a period (.) in the name which caused the bucket creation command to fail. Bucket names with dots require DNS verification and can cause compatibility issues. Fixed by choosing a bucket name using only lowercase letters numbers and dashes with no periods. - Key lesson: Stick to lowercase letters, numbers, and dashes when naming buckets to avoid naming rule violations.

Resources

GCP Cloud Storage CLI Docs: <https://cloud.google.com/storage/docs/discover-object-storage-gsutil>

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